

(or polar)

Graham Scan

Graham Scan(Q)

let p_0 be the point in Q with the minimum y-coordinate on the leftmost such point in case of a tie

//

A B C

consider A and B
then A is leftmost (rules)
so $p_0 = A$

let $\langle p_1, p_2, \dots, p_m \rangle$ be the remaining points in Q,

Sorted by polar angle in counterclockwise

order around p_0

(if more than one point has the same angle, remove all but the one that is farthest from p_0)

if $m < 2$

return "convex hull is empty"

else let S be an empty stack

push (p_0, S)

push (p_1, S)

$O(n \log n)$

Push (P_2, S)

for $i = 3$ to n

while the angle formed by points
Next-To-Top (S), Top (S) and P_i makes

a nonleft turn

Pop (S)

Push (P_i, S)

Return S .

Solve this

$A(0,0)$, $B(2,1)$

$E(5,1)$, $F(2,2)$

Sort by

0

1

2

3

4

5

A

B

E

F

C

D

$P_0 = A(0,0)$

Measure the angle from A to every point

B	$\tan^{-1}(\frac{1}{2}) = 26.57^\circ$	Sort
C	$\tan^{-1}(\frac{3}{1}) = 71.57^\circ$	E
D	$\tan^{-1}(\frac{9}{4}) = 65.9^\circ$	B
E	$\tan^{-1}(\frac{1}{5}) = 11.3099^\circ$	D
F	$\tan^{-1}(\frac{2}{2}) = 45^\circ$	F
		C

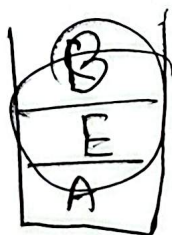
Remove duplicate angle

$$AD = \sqrt{(4-0)^2 + (4-0)^2} = 4$$

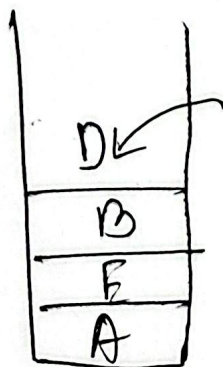
$$AF = \sqrt{(0-2)^2 + (0-2)^2} = \sqrt{8} = 2.83$$

remove F

Sorted order A E B D C



$$AE \times EB = \begin{vmatrix} 5 & 1 \\ -3 & 0 \end{vmatrix} = 4$$



$$EB \times BD = \begin{vmatrix} -3 & 0 \\ 2 & 3 \end{vmatrix}$$

$$= -9 \text{ (right to left)}$$

Pop B

C
D
E
A

$$EDC \times DC = \begin{vmatrix} 5 & -1 & 3 \\ -3 & 1 & 1 \end{vmatrix}$$

$$T_{2.25} = \left(\frac{1}{2}\right)^{10}$$

$$T_{2.17} = \left(\frac{1}{2}\right)^{10}$$

$$= 1 + 9 = 10 \text{ left turn}$$

final convex hull ~~A → E → D → C~~
~~A → E → D → C~~
~~A → E → D → C~~

sigma algorithm

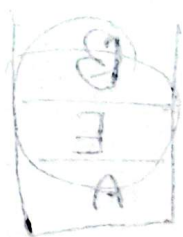
$$P = \sqrt{(0-1)^2 + (0-1)^2} = \sqrt{2}$$

$$E85 = 82 = \sqrt{(5-0)^2 + (5-0)^2} = 7.07$$

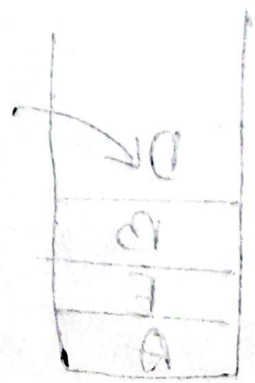
7.07

sorted order A E D C

$$A \times E = \begin{vmatrix} 1 & 2 \\ -3 & 0 \end{vmatrix} = 1$$



$$E \times D = \begin{vmatrix} 0 & 5 \\ 2 & 5 \end{vmatrix} = 0$$



(convex hull) →
 1000